

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway, EliteUS Software Solutions

Aug 2024 - Present | Piscataway, NJ

- Designed and deployed **GenAI** text/vision systems using **LLMs**, implementing **RAG** with semantic embeddings in vector databases, LangChain pipelines, custom prompt-orchestration to deliver high-accuracy responses.
- Built **analytics** systems for an e-commerce platform, developing **ETL** pipelines, customer segmentation models & **dashboards** while resolving data-quality issues, enabling accurate sales insights & real-time decision intelligence.
- Conducted **QA** on **analytics** pipelines & **dashboards** validating key commercial KPIs while using excel/sheets for **ad-hoc analysis**, scenario modeling, and cross-checks that enabled informed product, stakeholder decisions.
- Implemented **SQL**-driven recommender system pipelines & a text-to-SQL bot that analyzes customer purchase behavior & converts natural-language queries into insights using **analytics** and feature-engineered query logic.

Data Science Intern, Harvest Software Solutions, LLC

June 2023 - Sept 2023 | Jacksonville, Florida

- Led a **cross-functional** ads analytics initiative to build scalable data pipelines and monitoring systems, enabling reliable visibility into impression delivery, pacing behavior, and campaign performance across large-scale datasets.
- Engineered **ML**-driven workflows using H2O AutoML, XGBoost, and FastAPI NLP services to process large datasets, analyze sentiment/engagement patterns and boost predictive accuracy, spend efficiency for campaigns.
- Built **analytics** and evaluation systems combining **SQL** pipelines, forecasting models, and automated validation logic to assess engagement quality, optimize and support data-driven audience targeting for business teams.

Application Development Associate, Accenture Solutions Pvt. Ltd.

June 2021 - Aug 2022 | Hyderabad, India

Database Analyst, Regeneron Pharmaceuticals Inc. - Client Data Analysis & Warehousing

- Optimized **ETL** workflows and implemented **automated data pipelines in Python**, enhancing data processing efficiency.
- Performed large-scale **analytics** using SQL, AWS Athena, and Redshift on high-volume datasets, leveraging pandas for data cleaning and feature preparation, and visualized insights through interactive **dashboards** in Tableau and Power BI.
- Increased system reliability by 25% through optimization, performance tuning using **SQL Workbench**, while collaborating with **cross-functional teams** to document metric definitions, translate business needs into reporting solutions.

Research Lab, Andhra University

January 2021 - June 2021 | Visakhapatnam, India

Self Driving Cars Using Image Processing and Deep Learning - Researcher

- Developed a self driving car using vision models like CNNs, YOLO, Faster R-CNN, object classification and multi-class tracking, incorporating image preprocessing and multi-sensor fusion across camera, LiDAR, and radar streams.
- Built a high-fidelity autonomous driving simulation pipeline collaborating with engineers using Unity3D with a Python/Flask backend, supporting real-time inference, telemetry capture, and dynamic parameter tuning for system-level validation.

TECHNICAL SKILLS

Programming: Python, R, Scala, SQL. **Libraries:** Pytorch, Scikit-learn, Tensorflow, SparkML, Tableau, PowerBI.
Machine Learning: Ensemble Methods, Classification models, Gridsearching, Clustering, Regression Analysis, LightGBM.
Statistical Analysis: Causal Inference, Hypothesis testing, Time-series forecasting, stochastic process, Linear Regression.
Deep Learning: Neural Networks, Computer Vision, AI, NLP, BERT, GANs, LLMs, LLaMA, LSTM, OpenAI models.
Database & Cloud: Postgres, AWS, RDBMS, Data Warehousing, OpenAI, Anthropic, Gemini, GCP, Langchain, Pinecone.
Language Models: OpenAI, Azure, AWS Bedrock, Hugging Face Transformers, LLaMA, prompt tuning, RAG, Langchain.

EDUCATION

Rutgers University, Masters in Data Science

Sept 2022 - May 2024 | New Brunswick, NJ

Relevant Coursework: Statistics: Statistical Modeling and Computing, Stat Learning; Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. **Projects Overview:** [Portfolio](#).

Andhra University, B.Tech in Electrical and Electronics Engineering

2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Bayesian Classification - [Bayesian Networks, PyMC3, Informative Prior Distributions] Developed Bayesian logistic regression models using PyMC3 to predict diabetes status, incorporating informative priors. Evaluated performance with recall and f1-score metrics, comparing outcomes between uniform and normal distribution priors.

Language Identification - [Naive Bayes, Logistic Regression, BERT, GloVe embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

Auto Insurance Fraud Detection - [Python, Data Visualization, ML models - Random Forests, KNN, SVM, Decision Trees, XGBoost, Regression] Executed diverse ML strategies and time series analysis for auto insurance fraud, prioritizing reduced false negatives. Chose K-Nearest Neighbors (KNN) with an impressive 97% recall.

Thank you for considering my application. I'm available to start working immediately.